

LIU PAK WAI (PAUL)

Fullstack Developer | Backend Specialist 📍 Hong Kong | ☎ +852 9545 8756 | ✉ pwliuab@connect.ust.hk

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PROFESSIONAL SUMMARY

High-performing **Fullstack Developer** and **CityU Computer Science Master's graduate** with a strong foundation in **Computer Engineering** from **HKUST**. Previously a Senior Analysis Programmer at **YOHO**, specializing in scalable backend architecture, database optimization, and search engine implementation. Proven track record in optimizing system performance and handling large-scale data migrations.

EDUCATION

Master of Science in Computer Science

City University of Hong Kong (CityU) | 09/2022 – 12/2024

Bachelor of Engineering in Computer Engineering

The Hong Kong University of Science and Technology (HKUST) | 09/2018 – 06/2022

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), Node.js, Java, SQL, HTML5/CSS3, PHP, Python.
 - **Frameworks/Libraries:** Express.js, React.js, Next.js, React Native, Angular, Django
 - **Databases & Tools:** MySQL, MongoDB, Redis, Elasticsearch.
 - **Developer Tools:** Git, Visual Studio Code, Android Studio, Jupyter Notebook, Colab.
 - **Devops Tools:** Git Lab, Kubernetes, Docker, Google Cloud
 - **Languages:** Cantonese (Native), Mandarin (Native), English (Fluent)
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WORK EXPERIENCE

友和 **YOHO** | *Ecommerce*

Senior Analyst Programmer | 05/2026 – 09/2026

- **Financial Data Integrity:** Engineered complex **SQL validation scripts** and **advanced Excel models** to cross-reference transactional data; identified and rectified legacy inconsistencies to ensure accuracy in financial reporting.
- **High-Throughput Data Processing:** Optimized the generation of 1,000+ comprehensive monthly reports by implementing **Node.js streaming** and **bulk-create/fetch** techniques, successfully managing datasets with millions of individual order records.
- **Memory Management:** Eliminated system timeouts and memory overflows by replacing monolithic data fetches with a **chunked streaming architecture**, ensuring stable performance during peak billing cycles.

Senior Web Developer | 10/2023 – 05/2026

- **Search Engine Revolution:** Optimized platform search performance by **100x** by migrating legacy MySQL backend to an **Elasticsearch-driven architecture**.
- **Performance Engineering:** Resolved critical N+1 query issues through optimized data fetching strategies, significantly enhancing API response times.
- **Database Excellence:** Conducted a comprehensive audit of database indexes, achieving a **10% reduction in query latency**.
- **Data Integrity:** Restored system reliability by identifying and purging thousands of duplicate records and implementing stricter validation constraints in MySQL.

Web Developer | 09/2022 – 10/2023

- **Concurrency Management:** Eliminated race conditions by implementing **Redis-based distributed locking**, ensuring data integrity across concurrent processes.
- **Infrastructure Growth:** Architected and implemented a dedicated **Image Server**, reducing upload latency and improving overall system throughput.
- **Frontend Optimization:** Audited and optimized frontend performance, resulting in measurable increases in page load speeds and **Core Web Vitals**.
- **Refactoring:** Optimized **Sequelize ORM** performance by refactoring complex database relationships and streamlining model associations.
- **Data Scaling:** Normalized and migrated a **100k-row dataset**, improving efficiency and long-term scalability.

HKUST Path Advisor | *The University*

Front-end Developer (Part-time) | 02/2022 – 06/2022

- Maintained and scaled the existing React.js system used by the university community.

Aflame Tech Limited | *Vendor Company*

Software Developer Intern (Part-time) | 09/2021 – 06/2022

- Built a cross-platform mobile application from scratch using **React Native**.
- Developed backend services using **Node.js** and **MongoDB** for seamless data persistence.
- Created an internal administrative dashboard using **React Admin** to streamline operations.